

Tutoring - First Law, Second Law

Q. An inventor claims that his engine has the following specifications:

Temperature limits: 750°C - 25°C

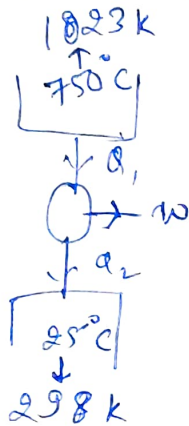
Power developed: 75 kW

Fuel burned per hour: 3.9 kg

Heating value of the fuel: 74500 kJ/kg

State whether his claim is valid or not.

Solⁿ.



$$\text{power} = 75\text{ kW} = 75 \frac{\text{kJ}}{\text{sec}}$$

$$\dot{m} = 3.9\text{ kg/hr}$$

$$C_v = 74500\text{ kJ/kg}$$

$$Q_1 = 3.9 \frac{\text{kg}}{\text{hr}} \times 74500 \frac{\text{kJ}}{\text{kg}}$$

$$Q_1 = \frac{3.9 \times 74500}{3600} \frac{\text{kJ}}{\text{sec}}$$

$$Q_1 = 80.70\text{ kW}$$

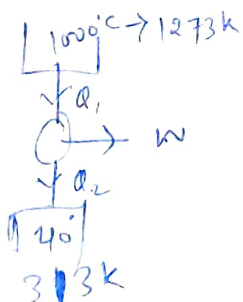
$$\eta_{\text{Carnot}} = 1 - \frac{298}{1023} = \underline{70.86\%}$$

$$\eta_{\text{th}} = \frac{75}{80.70} = 92.93\%$$

Claims
- not valid

Q. A cyclic heat engine operates b/w a source temp^r of 100°C & a sink temp^r of 40°C . Find the least rate of heat rejection per kW net output of the engine?

Solⁿ.



$$W = 1\text{ kW}$$

$$\frac{W}{Q_2} = \frac{T_1 - T_2}{T_1}$$

$$\eta_{\text{rev}} = 1 - \frac{313}{373} = \frac{W}{Q_1}$$

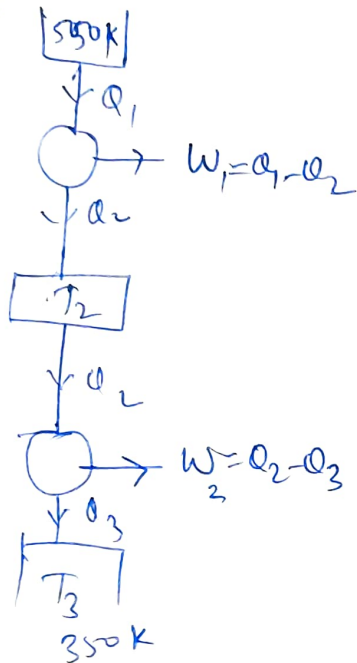
$$Q_1 = 1.3260\text{ kW}$$

$$Q_2 = Q_1 - W = 0.326\text{ kW}$$

Q. Two Carnot engines work in series b/w the source & sink temp. of 550 K & 350 K. If both engines develop equal power; determine the intermediate temp.

9/5/11

Soln.



$$W_1 = W_2$$

$$\Rightarrow Q_1 - Q_2 = Q_2 - Q_3$$

$$\frac{Q_1}{T_1} = \frac{Q_2}{T_2} = \frac{Q_3}{T_3}$$

$$\Rightarrow Q_2 = \frac{T_2}{T_1} \cdot Q_1 =$$

$$Q_3 = \frac{T_3}{T_1} \cdot Q_1$$

$$\Rightarrow Q_1 - \frac{T_2}{T_1} Q_1 = \frac{T_2}{T_1} Q_1 - \frac{T_3}{T_1} Q_1$$

$$\Rightarrow 2 \frac{T_2}{T_1} Q_1 = Q_1 + \frac{T_3}{T_1} Q_1$$

$$T_2 = \left(1 + \frac{350}{550} \right) \times \frac{550}{2}$$

$$T_2 = 400 \text{ K}$$

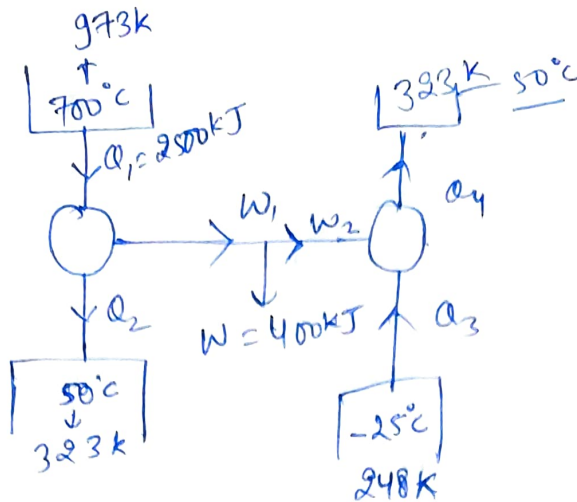
Q. A reversible heat engine operates b/w two reservoirs at temperatures of 70°C & 50°C. The engine drives a reversible refrigerator which operates b/w reservoirs at temp. of 50°C & -25°C. The heat transfer to the engine is 2500 kJ and the net work output of the combined engine refrigerator plant is 400 kJ.

Soln.

(i) Determine the heat transfer to the refrigerant & the net heat transfer to the reservoir at 50°C;

(ii) Reconsider (i) given that the efficiency of the heat engine and the COP of the refrigerator are each 45% of their maximum possible values.

Solⁿ



(i) $Q_2 + Q_4 = ?$
 $Q_3 = ?$

$W = W_1 - W_2 = 400 \text{ kJ}$

$\frac{Q_1}{T_1} = \frac{Q_2}{T_2} \Rightarrow Q_2 = \frac{Q_1 T_2}{T_1} = \frac{2500 \times 323}{973} = 820 \text{ kJ}$

$\Rightarrow Q_2 = Q_1 \times \frac{T_2}{T_1} = 2500 \times \frac{323}{973} = 820 \text{ kJ}$
 $W_1 = Q_1 - Q_2 = 2500 - 820 = 1680 \text{ kJ}$

$W_2 = 1680 - 400 = 1280 \text{ kJ}$

$W_2 + Q_3 = Q_4$

$W_2 = Q_4 - Q_3 \Rightarrow 1280 = Q_4 - Q_3$

$\text{COP} = \frac{W_2}{Q_3} = \frac{T_4 - T_3}{T_3} \Rightarrow \frac{1280}{Q_3} = \frac{323 - 248}{248}$

$Q_3 = 4199 \text{ kJ}$

$Q_3 = \frac{Q_4}{323} \times 248$
 $Q_3 = 0.7678 Q_4$

$Q_4 = W_2 + Q_3$

$1280 = Q_4 - 0.7678 Q_4$
 $Q_4 = 5469 \text{ kJ}$

$\Rightarrow 1280 = Q_4 - 0.7678 Q_4$

$\Rightarrow Q_4 = 4853.57 \text{ kJ}$

$Q_3 = 0.7678 \times Q_4 = 3726.57$

$Q_3 = 4199 \text{ kJ}$

$Q_2 + Q_4 = 820 + 5469$

$Q_2 + Q_4 = 6299 \text{ kJ}$

(ii) $\eta = \frac{W_1}{Q_1} = 0.45$

$\eta_{H-S} = 0.459_{max} = \frac{W_1}{Q_1} = 0.45 \left(1 - \frac{T_2}{T_1} \right) \Rightarrow W_1 = 751.54 \text{ kJ}$

$W_2 = W_1 - 400 = 351.54 \text{ kJ}$, $Q_2 = Q_1 - W_1 = 2500 - 751.54 = 1748.46 \text{ kJ}$

$\text{COP} = \frac{Q_3}{W_2} \Rightarrow \frac{Q_3}{W_2} = 0.45 \left[\frac{T_3}{T_4 - T_3} \right] \Rightarrow Q_3 = 523.09 \text{ kJ}$

$W_2 + Q_3 = Q_4 \Rightarrow Q_4 = 351.54 + 523.09 = 874.63 \text{ kJ}$
 $Q_2 + Q_4 = 1748.46 + 874.63 = 2623.09 \text{ kJ}$

$Q_3 = 523.09 \text{ kJ}$

$Q_4 = 874.63$
 $Q_2 + Q_4 = 1748.46 + 874.63 = 2623.09 \text{ kJ}$

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